THE COTANGENT COMPLEX OF A MORPHISM OF SCHEMES IN THE CONSTANT BASE CASE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file cotangent.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The statement displays a chain of three equalities; the proof justifies the last two first and then identifies the object with the cotangent complex of the ring map.

Editorial note. The three references of the counted spans are recorded as non-load-bearing navigation pointers; the argument that consumes them stays inside the counted proof span.

Editorial note. Preamble: the retained preamble of this chapter (sources/stacks-cotangent/retained-preamble.tex, lines 1-63), containing the Xy-pic setup, the comment environments, the theorem declarations and the macros including \NL.

Allowed standard prerequisites (editorial).

- The computation of the lower shriek as sheafification of an explicit presheaf complex. Remark remark-compute-L-pi-shriek, used at proof lines 4362-4374. Describes the object whose equality is claimed.
- The section-level map identifying the complex with the ring-level cotangent complex. Remark remark-map-sections-over-U, used at proof lines 4375-4379. Identifies the complex as computing the cotangent complex of Lambda -> O_X.
- The description of the cotangent complex of a scheme over a ring. Lemma lemma-scheme-over-ring, used at proof lines 4379-4380. Concludes the identification with L_{X/Lambda}.
- Flatness of the module of differentials for the objects of the category. used at proof lines 4354-4359. Gives the second and third equalities of the displayed chain.

Source reference. Original lemma 08V6: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 11224–11224. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original remark 08VB: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 11229–11229. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original remark 08SX: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 11145–11145. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Date: stacks-project:cotangent.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Lemma 0.2. *In the situation above there is a canonical isomorphism*

$$L_{X/\Lambda} = L\pi_!(Li^*\Omega_{\mathcal{O}/\underline{\Lambda}}) = L\pi_!(i^*\Omega_{\mathcal{O}/\underline{\Lambda}}) = L\pi_!(\Omega_{\mathcal{O}/\underline{\Lambda}} \otimes_{\mathcal{O}} \underline{\mathcal{O}}_X)$$

in $D(\mathcal{O}_X)$.

Proof. We first observe that for any object $(U \to \mathbf{A})$ of $\mathcal{C}_{X/\Lambda}$ the value of the sheaf $\mathcal{O}$ is a polynomial algebra over Λ. Hence $\Omega_{\mathcal{O}/\underline{\Lambda}}$ is a flat $\mathcal{O}$-module and we conclude the second and third equalities of the statement of the lemma hold.

By Remark 08VB the object $L\pi_!(\Omega_{\mathcal{O}/\underline{\Lambda}} \otimes_{\mathcal{O}} \underline{\mathcal{O}}_X)$ is computed as the sheafification of the complex of presheaves

$$U \mapsto \left(\Omega_{\mathcal{O}/\underline{\Lambda}} \otimes_{\mathcal{O}} \underline{\mathcal{O}}_X\right)(\mathbf{A}_{\bullet,U}) = \Omega_{P_{\bullet,U}/\Lambda} \otimes_{P_{\bullet,U}} \mathcal{O}_X(U) = L_{\mathcal{O}_X(U)/\Lambda}$$

using notation as in Remark 08VB. Now Remark 08SX shows that $L\pi_!(\Omega_{\mathcal{O}/\underline{\Lambda}} \otimes_{\mathcal{O}} \underline{\mathcal{O}}_X)$ computes the cotangent complex of the map of rings $\underline{\Lambda} \to \mathcal{O}_X$ on X. This is what we want by Lemma 08V6. $\square$